

Pedagogy: Intelligence, Aptitude, Individual Differences and Sociometry

शिक्षाशास्त्र: बुद्धि, अभिक्षमता, वैयक्तिक भिन्नताएँ और समाजमिति

Intelligence, Aptitude, Individual Differences and Sociometry

What UPSC asks on this - Who gave which theory: Spearman's g, Thurstone's 7 primary abilities, Guilford's 180-cell model, Gardner's 8 intelligences, Sternberg's triarchic theory, the PASS model, Cattell's fluid and crystallised intelligence. - IQ history and arithmetic: Binet 1905, Stern's IQ 1912, Terman 1916, deviation IQ with mean 100 and SD 15, the NCERT IQ bands and their percentages. - Types of tests: individual vs group, verbal vs performance, culture-fair; Indian tests such as Bhatia's Battery. - Aptitude vs intelligence vs achievement vs interest, and the eight DAT subtests. - Individual differences: twin-study correlations, causes, classroom provisions, and the definitions of giftedness and intellectual disability. - Sociometry: Moreno, sociogram vocabulary (star, isolate, mutual pair, clique) and what the technique can and cannot tell a teacher.

1. Why a Vice Principal needs this

In one line: every placement, remedial group, CWSN referral and counselling decision in a school rests on these four ideas.

NEP 2020's first fundamental principle is "recognizing, identifying, and fostering the unique capabilities of each student". RTE Section **29(2)(e)** requires "learning through activities, discovery and exploration in a child friendly and child-centered manner". Both assume the teacher can tell intelligence from aptitude, measure each sensibly, and read the social map of a classroom. CRT questions on this topic follow the vocabulary of the NCERT Class XII Psychology chapter *Variations in Psychological Attributes* and standard B.Ed texts, so the facts below use that vocabulary.

2. Individual differences

In one line: no two learners are alike; differences exist between persons (inter-individual) and within one person across traits (intra-individual).

NCERT defines individual differences as the "distinctiveness and variations among people's characteristics and behaviour patterns". The opposing view is **situationism**: behaviour is shaped more by the situation than by stable traits.

Historical markers:

- **Francis Galton** (*Hereditary Genius*, 1869; anthropometric laboratory, 1884) began the scientific study of individual differences.

- **James McKeen Cattell** coined the term "mental tests" (1890).
- **Alfred Binet** turned the idea into a school tool in 1905, when the French education authorities asked him to identify children who needed special teaching.

Two types:

- **Inter-individual:** Asha reads faster than Rohan.
- **Intra-individual:** Asha is strong in language but weak in mathematics. Guidance works on this profile, not on one score.

Pupils differ in physical growth, intelligence, aptitude, interest, attitude, personality, achievement, learning pace and style, motivation, gender, language and socio-cultural background, and special needs.

Causes (the standard exam list): **heredity, environment** (home, school, peers, media), **age and maturation, sex, socio-economic status, education and training, culture**. The B.Ed formula: *heredity sets the limits; environment decides how far within them a child goes.*

Heredity and environment: the twin evidence

NCERT gives correlations of intelligence between relatives. Memorise the ladder:

Pair	Correlation
Identical twins reared together	0.90
Identical twins separated early	0.72
Fraternal twins reared together	0.60
Siblings reared together	0.50
Siblings reared apart	0.25

Reading it: the same genes in different homes (0.72) still correlate more than different genes in the same home (0.60), so heredity matters. But 0.90 against 0.72 shows a shared home adds something too. Adopted children's intelligence resembles their **biological** parents more than their adoptive parents, yet children moved from deprived to enriched homes gain IQ points. NCERT's conclusion: intelligence is a product of the **interaction of heredity and environment**.

Heredity gives a range of possibilities; environment decides where in that range the child lands.

What a school does about individual differences

- **Know the learner:** baseline tests, cumulative record card, anecdotal record, interest inventory, sociogram, home visit.
- **Differentiated instruction:** same learning outcome, different routes, pace and products. **Universal Design for Learning (UDL)** builds the variety in from the start instead of retro-fitting it.
- **Grouping:** ability grouping for a specific skill (reading levels), mixed-ability grouping for projects. Groups stay flexible, never permanent labels.

- **Remedial and enrichment:** remedial classes for those below level; supplementary material, clubs and Olympiads for the advanced (NEP 2020, Chapter 4, under "Support for Gifted Students/Students with Special Talents").
- **Individualised Education Programme (IEP)** for CWSN, prepared with the special educator. The **RPwD Act 2016** lists **21** specified disabilities, including intellectual disability, specific learning disabilities and autism spectrum disorder, and makes early detection of specific learning disabilities a duty of schools (**s.16**).
- **Guidance:** Delhi government schools have **Educational and Vocational Guidance Counsellors (EVGCs)**; the VP routes referrals to them.

Delhi example: a Sarvodaya Vidyalaya VP reviews Class 9 results. Instead of one "weak section", the staff run reading-level groups in Hindi and English periods, keep mixed groups for science projects, send three repeat-failures to the EVGC for aptitude testing, and refer a child with persistent letter reversals to the special educator for a specific learning disability screening. That is individual differences applied.

3. Intelligence: what it is

In one line: intelligence is the global capacity to understand, think rationally and deal effectively with the environment; definitions fall into three families.

- **Ability to learn:** Buckingham, "intelligence is the ability to learn".

- **Ability to adapt:** Stern, the "general capacity to adjust to new situations"; Piaget, adaptation through assimilation and accommodation.
- **Ability to think abstractly:** Terman, "the ability to carry on abstract thinking"; Binet stressed judgment and reasoning.
- **Composite: Wechsler**, "the global and aggregate capacity of an individual to act purposefully, to think rationally and to deal effectively with his environment". This is the definition most textbooks quote.

Thorndike added that intelligence shows in three kinds of material: **abstract** (words, symbols), **concrete or mechanical** (objects) and **social** (people).

NCERT contrasts Western **technological intelligence** (speed, abstract reasoning, individual achievement) with the Indian notion of **buddhi** or **integral intelligence**, which includes **cognitive capacity, social competence, emotional competence and entrepreneurial competence**.

4. Theories of intelligence

In one line: theories move from one factor, to many factors, to processes; for each, know the person, the number and the key word.

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`mermaid flowchart TD
  T[Theories of intelligence] --> P[Psychometric factor theories]
  T --> I[Information processing theories]
  T --> M[Multiple intelligences]
  P --> S[Spearman g and

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s] P --> U[Thurstone 7 PMAs] I --> R[Sternberg triarchic] M --> G[Gardner 8 intelligences] ``

Psychometric (factor) theories

- **Unifactor:** Binet and Terman treated intelligence as one general capacity. Trap: "Binet proposed g" is wrong; g is Spearman's term.
- **Spearman's two-factor theory (1904; *The Abilities of Man*, 1927):** every mental task needs a **general factor g** shared by all tasks plus a **specific factor s** unique to that task. Found through **factor analysis** of test correlations. Why it matters: g is the justification for a single IQ score.
- **Thorndike's multifactor theory:** there is no g; intelligence is a bundle of many independent abilities. Four attributes: **level, range, area, speed**. (The "sampling theory" is Godfrey Thomson's, a separate theory; do not confuse the two.)
- **Thurstone's group-factor theory (1938):** between g and s lie group factors. He named **7 Primary Mental Abilities:** verbal comprehension (V), word fluency (W), number (N), space (S), associative memory (M), perceptual speed (P), reasoning (R). Mnemonic: **"Very Wise Numbers Serve Many Pupils' Reasoning."**
- **Guilford's Structure of Intellect:** three dimensions, **operations × contents × products**. Originally **120** cells ($5 \times 4 \times 6$), later 150, and finally **180** ($6 \text{ operations} \times 5 \text{ contents} \times 6 \text{ products}$), the figure NCERT gives. Operations include **convergent** and

divergent production; divergent production is the root of creativity testing.

- **Hierarchical models (Burt, Vernon):** g at the top, then major group factors **v:ed** (verbal-educational) and **k:m** (spatial-mechanical), then minor group factors, then specifics.
- **Cattell (with Horn): fluid and crystallised intelligence.** **Fluid (gf)** is raw reasoning, relatively culture-free, peaks in early adulthood and **declines**; **crystallised (gc)** is accumulated knowledge and skill and **grows** with schooling and age. Non-verbal matrix tests tap gf; vocabulary taps gc.

Information-processing theories

- **Sternberg's triarchic theory (1985):** three aspects. **Componential (analytical):** metacomponents plan, performance components execute, knowledge-acquisition components learn. **Experiential (creative):** coping with novelty and automatising the routine. **Contextual (practical):** adapting to, shaping or selecting environments. Sternberg later built these into his theory of "successful intelligence". Contrast: Sternberg **three**, Gardner **eight**.
- **PASS model (J.P. Das, Jack Naglieri and John Kirby, 1994):** built on Luria's three functional units of the brain. **Planning, Attention-arousal, Simultaneous** processing (grasping the whole, as in Raven's matrices) and **Successive** processing (serial order, as in digit span). Das and Naglieri's **Cognitive**

Assessment System (CAS) measures these for ages **5 to 18** (NCERT).

Gardner's theory of multiple intelligences

Howard Gardner, *Frames of Mind* (1983), rejected a single g and proposed relatively independent intelligences, each with its own brain basis and developmental path. There were **seven** at first; **naturalistic** was added in the 1990s, and NCERT lists **eight**: linguistic, logical-mathematical, spatial, musical, bodily-kinaesthetic, interpersonal, intrapersonal, naturalistic. Gardner has discussed a ninth, **existential**, without formally adding it. His evidence came from prodigies, brain-damaged patients and cross-cultural study, **not factor analysis** (a common trap).

Mnemonic: "**SLIM BLIN**": Spatial, Linguistic, Interpersonal, Musical, Bodily-kinaesthetic, Logical, Intrapersonal, Naturalistic.

School meaning: the Holistic Progress Card developed by NCERT and PARAKH assesses many domains, not only scholastic ones. A child weak on logical-mathematical items may be strong on musical or interpersonal ones. NEP 2020's "no hard separation" between arts, sciences, sports and vocational learning follows the same logic.

Emotional intelligence

Coined by **Salovey and Mayer (1990)**: the ability to perceive, use, understand and manage emotions in oneself and others. Popularised by **Daniel Goleman (1995)**; it is expressed as **EQ** (emotional quotient), on

the model of IQ. Goleman's five components: self-awareness, self-regulation, motivation, empathy, social skills. NCERT calls emotional intelligence "the feeling side of intelligence". School link: social-emotional learning, peer mediation and counselling build EQ, which is linked with better adjustment and is a quality a high IQ alone does not guarantee.

5. Measuring intelligence

In one line: Binet started it, Stern gave the formula, Terman multiplied by 100, Wechsler replaced the formula with the normal curve.

``mermaid timeline 1905 : Binet-Simon scale, Paris 1908 : Mental age introduced 1912 : Stern coins IQ 1916 : Terman Stanford-Binet scale 1917 : Army Alpha and Beta 1938 : Raven's Progressive Matrices 1939 : Wechsler-Bellevue scale 1955 : Bhatia's Battery, India ``

Ratio IQ and its arithmetic

Mental age (MA) is the age level at which a child performs: a 7-year-old who passes the items most 9-year-olds pass has MA 9. **Stern (1912)** proposed the mental quotient MA/CA ; **Terman (1916)** multiplied it by 100 to remove the decimal.

$$IQ = (MA \div CA) \times 100$$

Worked examples (convert to months when ages include months):

1. CA 8 years, MA 10 years. $IQ = 10/8 \times 100 = 125$.

2. CA 12 years 6 months = 150 months; MA 10 years = 120 months. $IQ = 120/150 \times 100 = 80$.
3. IQ 120, CA 10 years. $MA = IQ \times CA \div 100 = 120 \times 10 \div 100 = 12 \text{ years}$.
4. Two pupils both have MA 12. One is 10 years old (IQ **120**), the other 15 (IQ **80**). Same mental age, very different IQ: IQ is relative to age.
5. CA 9 years 4 months = 112 months; MA also 112 months. $IQ = 100$. MA equal to CA always gives exactly 100.

Shortcut: the percentage by which MA leads CA is the IQ excess over 100 (MA 10% ahead gives IQ 110; MA 20% behind gives IQ 80).

Why ratio IQ died: mental age stops rising in a regular way around **15 to 16 years**, so MA/CA shrinks for every adult. **Wechsler** introduced the **deviation IQ**: the raw score is placed on a normal curve with **mean 100 and SD 15**. All modern scales use this.

Deviation IQ: reading the normal curve

Under a normal curve about 68% of people lie within 1 SD of the mean and about 95% within 2 SD.

1. IQ 115 = +1 SD. Percentile $\approx 50 + 34 = 84\text{th}$.
2. IQ 130 = +2 SD. Percentile $\approx 50 + 34 + 13.5 \approx 97.7\text{th}$: only about **2.2%** score higher.
3. IQ 85 = -1 SD. Percentile $\approx 16\text{th}$; about 16% of pupils score below 85.

NCERT's classification (Wechsler-based):

IQ range	Label	% of people
130 and above	Very superior	2.2
120–130	Superior	6.7
110–120	High average	16.1
90–110	Average	50.0
80–90	Low average	16.1
70–80	Borderline	6.7
Below 70	Intellectually disabled	2.2

Pattern to memorise: **2.2, 6.7, 16.1, 50, 16.1, 6.7, 2.2**, symmetrical about 100.

Types of tests

- **Individual vs group:** Binet and Wechsler scales are individual (one trained examiner, one child). The **Army Alpha** (verbal) and **Army Beta** (non-verbal, for illiterate and non-English recruits) of **1917** were the first large-scale group tests. Group tests are cheaper; individual tests allow richer observation.
- **Verbal vs non-verbal vs performance:** verbal tests need reading; non-verbal tests use figures (Raven's Progressive Matrices); performance tests use objects (block design, pass-along, form boards) and suit children with language or hearing difficulties.
- **Culture-fair vs culture-biased:** tests built on language and school knowledge favour the culture that built them. **Raven's Progressive Matrices (J.C. Raven, 1938)** and **Cattell's Culture Fair Test** try to reduce this. The accepted view: a fully culture-free

test is not possible, only culture-reduced (culture-fair) ones.

- **Wechsler family:** Wechsler-Bellevue (1939), **WISC** for children (1949), **WAIS** for adults (1955), **WPPSI** for pre-schoolers (1967). The original Wechsler scales gave separate verbal and performance IQs, which Binet scales did not (current editions report several index scores instead).
- **Indian tests: Bhatia's Battery of Performance Tests of Intelligence (C.M. Bhatia, 1955)**, five performance subtests including Koh's Block Design and the Alexander Pass-along test; **Malin's Intelligence Scale for Indian Children (MISIC)**, an Indian adaptation of the WISC.

Cautions NCERT wants you to know

An IQ is a sample of behaviour on one day, affected by language, test-wiseness, anxiety and culture. It must not label a child permanently, must be read with school performance and observation, and predicts school success only moderately. A VP reading a test report should ask which test, which norms and who administered it.

6. The extremes: intellectual disability, giftedness, creativity

In one line: both tails of the curve need a plan; the cut-offs are 70 and 130.

Intellectual disability

The definition quoted in NCERT (from the American Association on Mental Deficiency, now the **American**

Association on Intellectual and Developmental Disabilities, AAIDD) needs **three conditions together**: significantly sub-average intellectual functioning (**IQ about 70 or below**), **deficits in adaptive behaviour** (self-care, communication, social skills), and onset **during the developmental period** (NCERT: 0 to 18 years; the AAIDD's 2021 definition extends this to before age 22). A low IQ alone is not enough.

Levels by approximate IQ (NCERT): **mild about 55–70** (educable; the majority), **moderate 35–40 to 50–55** (trainable), **severe 20–25 to 35–40, profound below 20–25**. Under the **RPwD Act 2016**, intellectual disability is a specified disability; a child with a **benchmark disability (40% or more)** aged **6 to 18** has the right to free education in a neighbourhood or special school (**s.31**).

School practice: IEP, resource room, special educator, examination relaxations under board rules, and emphasis on adaptive and vocational skills.

Slow learners and under-achievers

A **slow learner** has an IQ roughly in the borderline and low-average bands (about **70 to 90**), learns at a slower pace and needs concrete, repeated, small-step teaching. This is not a disability. An **under-achiever** scores well below what measured ability predicts; the cause is usually motivational, emotional or environmental, so the response is counselling, not remediation alone. Confusing the two is a classic trap.

Giftedness and talent

NCERT: **gifted** means exceptional general ability shown as superior performance across many areas; the conventional IQ cut-off is **130 and above**. **Talented** means remarkable ability in a specific field (music, mathematics, sport). **Terman's** longitudinal study of about 1,500 gifted children (from 1921) selected IQs around 140 and above and found the gifted were, as a group, healthy and well adjusted. **Renzulli's three-ring conception**: giftedness is the overlap of **above-average ability, creativity and task commitment**.

Provision: acceleration (moving faster), enrichment (going deeper), grouping, mentoring. NEP 2020 lists supplementary enrichment material, **topic-centred and project-based clubs and circles** (Science, Maths, Chess, Debate, Eco and others), **Olympiads and competitions**, and asks **NCERT and NCTE** to develop guidelines for the education of gifted children. In Delhi, the **Rajkiya Pratibha Vikas Vidyalayas (RPVV)** and **Schools of Specialised Excellence (SoSE)** admit students through entrance or aptitude tests for this purpose.

Creativity

Guilford separated **convergent** thinking (one right answer, what IQ tests reward) from **divergent** thinking (many possible answers), making divergent production the core of creativity. The **Torrance Tests of Creative Thinking** score **fluency** (how many ideas), **flexibility** (how many categories), **originality** (how rare) and **elaboration** (how detailed). **Wallas's four stages** of the

creative process: preparation, incubation, illumination, verification. Indian tests: **Baquer Mehdi's** Verbal and Non-verbal Tests of Creative Thinking and **Passi's** Tests of Creativity.

Relation with intelligence (NCERT): positive but moderate. A minimum intelligence is needed for creativity (the **threshold** is usually put near IQ 120), but above it a higher IQ does not guarantee more creativity. A high-IQ child is not necessarily creative, and a creative child may have an average IQ.

7. Aptitude and interest

In one line: intelligence is general capacity, aptitude is specific potential, achievement is what has been learnt, interest is what one likes.

Aptitude (NCERT) is "a combination of characteristics indicative of an individual's capacity to acquire some specific knowledge or skill after training". It is **potential**, measured before training and used for **prediction**. **Aptitude plus interest** is what counsellors look for: aptitude without interest produces a bored engineer; interest without aptitude produces frustration.

Construct	Question it answers	Typical tool
Intelligence	How well can you learn in general?	Wechsler, Stanford-Binet, Raven's
Aptitude	How well could you learn this specific skill?	DAT, GATB, mechanical or clerical tests
Achievement	What have you already learnt?	Unit test, board exam, PARAKH survey
Interest	What do you like doing?	Strong, Kuder, Thurstone Interest Schedule

Aptitude tests

Two forms (NCERT): **independent (specialised)** tests for one field, such as mechanical, clerical, musical (Seashore Measures of Musical Talents) or teaching aptitude; and **multiple (generalised)** aptitude **batteries**.

- **Differential Aptitude Tests (DAT)**, Bennett, Seashore and Wesman, **1947: 8 subtests**: Verbal Reasoning, Numerical Ability (NCERT: Numerical Reasoning), Abstract Reasoning, Clerical Speed and Accuracy, Mechanical Reasoning, Space Relations, Spelling, Language Usage. The most used battery in schools. **J.M. Ojha** developed the Indian adaptation (NCERT). Memory picture: a report card with Verbal, Numerical, Abstract on top; Clerical, Mechanical, Space in the middle; Spelling and Language at the bottom.
- **General Aptitude Test Battery (GATB)**: developed by the US Employment Service; measures **nine**

aptitudes for job matching.

- **Armed Services Vocational Aptitude Battery (ASVAB):** US military selection.
- India also uses **David's Battery of Differential Abilities (DBDA)**, an Indian multi-aptitude battery.

Names can mislead. NCERT's National Talent Search Examination had a **Mental Ability Test (MAT)** and a **Scholastic Aptitude Test (SAT)**; the MAT measured reasoning, while the SAT tested school subjects and so worked more like an achievement test.

Interest inventories

Strong Vocational Interest Blank (1927), Kuder Preference Record, Thurstone Interest Schedule; in India, **Chatterji's Non-Language Preference Record.** Interest is measured by self-report, so it can be faked and changes with age; aptitude is measured by performance.

Delhi example: at Class 10 stream choice, the EVGC administers a DAT-type battery and an interest inventory, then counsels the student and parent. The VP schedules the testing, keeps the records confidential and ensures the results inform the choice rather than dictate it.

8. Sociometry

In one line: sociometry maps who chooses whom in a group; J.L. Moreno created it and the sociogram displays it.

Jacob L. Moreno, a Romanian-born psychiatrist, published *Who Shall Survive?* in **1934** and founded sociometry (he also founded **psychodrama**). **Helen Hall Jennings** worked with him on the early school studies. The technique measures the **informal structure** of a group: the attractions, rejections and indifference that the formal seating plan hides.

Procedure

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`mermaid flowchart TD
  A[Fix a real criterion] --> B[Each pupil names 3 choices]
  B --> C[Tabulate in socio-matrix]
  C --> D[Draw sociogram]
  D --> E[Spot stars and isolates]
  E --> F[Act on seating and groups]`
```

1. **Criterion:** a real, specific activity ("Which three classmates would you like to sit with on the school trip?"). Different criteria give different maps; a work criterion and a play criterion rarely coincide.
2. **Choices:** usually **three** positive nominations. Some versions add negative nominations ("would not like to"), which raise ethical concerns in schools.
3. **Confidentiality**, and a promise that the results will actually be used. Moreno insisted the test be real, not hypothetical.
4. **Socio-matrix:** rows are choosers, columns are chosen; column totals give each pupil's **sociometric status** (choices received).
5. **Sociogram:** a diagram with **circles for girls and triangles for boys**, arrows for choices and double arrows for mutual choices. **Northway's target sociogram** places the most-chosen pupil at the

centre of concentric circles and the least-chosen at the edge.

Vocabulary (the exam's favourite)

- **Star:** receives the most choices.
- **Isolate:** receives **no** choices (and usually makes few).
- **Neglectee:** receives few or no choices but no rejections; simply overlooked.
- **Rejectee:** receives negative nominations.
- **Mutual pair (reciprocal choice):** A chooses B and B chooses A.
- **Chain:** A chooses B, B chooses C, C chooses D; one-way links.
- **Triangle:** three pupils who all choose one another.
- **Clique:** a closed sub-group whose members choose only each other.
- **Island:** a small sub-group with no link to the main group.

Coie and Dodge gave five status types still used in research: **popular, rejected, neglected, controversial** (many likes and many dislikes) and **average**. Rejected children carry the highest risk of later problems; neglected children often improve on their own when the group changes.

Worked example

Six pupils each name **two** classmates they would like to work with on a science project (12 choices in all).

Count the choices received:

Pupil	Chosen by	Received
Asha	Bina, Charu, Dev, Esha, Farhan	5
Bina	Asha, Charu	2
Charu	Asha, Bina	2
Dev	Esha, Farhan	2
Esha	Dev	1
Farhan	nobody	0

Check: $5 + 2 + 2 + 2 + 1 + 0 = 12$, the total number of choices made.

Reading: **Asha is the star** (5 of a possible 5). **Farhan is an isolate** (0 received, although he chose Asha and Dev). Asha–Bina, Asha–Charu, Bina–Charu and Dev–Esha are **mutual pairs**; Asha, Bina and Charu form a **triangle** that chooses only within itself, a small clique. Action: place Farhan in a group with one of the pupils he chose (Asha or Dev), do not put the whole triangle together or the clique hardens, and ask the EVGC to check whether Farhan is new, shy or being excluded.

Uses in school

Seating plans and group formation; identifying isolates and rejectees for guidance; measuring class **cohesiveness** (the proportion of mutual choices); repeating the test to check the effect of an intervention; choosing peer tutors and monitors who are actually accepted; studying leadership. B.Ed assessment courses (such as IGNOU BES-127) treat sociometry as a **non-testing technique of**

assessment, alongside observation, anecdotal records, rating scales and checklists.

Limitations

Results hold only for the **criterion and time** used; they show **who** is chosen, not **why**; young children's choices are unstable; negative nominations can hurt; absent pupils distort the map; and a star at "play" may be an isolate at "study". Sociometry **describes** a group; it does not **explain** it, so follow it with observation and interview.

Related techniques

Guess-who technique (Hartshorne and May): pupils name classmates who fit short descriptions ("someone who is always ready to help"). **Bogardus Social Distance Scale (1925)**: degrees of acceptance of a social group. **Peer rating** (rate every classmate on a scale) avoids the harshness of nominations and gives a score for everyone.

Common traps

- **Stern vs Terman vs Binet**: Binet gave the first scale (1905) and mental age (1908); Stern gave the IQ ratio (1912); Terman multiplied by 100 in the Stanford-Binet (1916); Wechsler gave the deviation IQ.
- **g is Spearman's**, not Binet's or Thorndike's. Thorndike denied g.
- **Thurstone 7, Gardner 8, Sternberg 3, Guilford 180 (originally 120), PASS 4.**

- **Gardner did not use factor analysis;** Spearman and Thurstone did.
- **Fluid declines with age, crystallised grows.**
Options often swap them.
- **IQ 70 and 130** are the NCERT cut-offs, with **2.2%** at each tail. "Superior" is 120–130; "very superior" is 130 and above.
- **Intellectual disability needs three conditions,** not a low IQ alone.
- **Slow learner, under-achiever and learning disability are three different things.**
- **Aptitude is potential; achievement is attainment.**
A board exam is an achievement test.
- **DAT has 8 subtests;** GATB has 9 aptitudes. Musical aptitude is not in the DAT.
- **Moreno** founded sociometry; Jennings was his collaborator; Northway gave the target sociogram.
- **Isolate receives no choices; rejectee receives negative choices; star receives the most.**
- **Sociometry is a non-testing, descriptive technique;** it is not a personality test and gives no causes.
- **Identical twins apart (0.72) correlate higher than fraternal twins together (0.60).** Options reverse these.

Remember for the exam

1. Binet-Simon scale 1905; mental age 1908; Stern's IQ 1912; Terman's Stanford-Binet 1916.

2. $IQ = MA \div CA \times 100$; deviation IQ has mean 100 and SD 15.
3. NCERT bands: 130 and above very superior (2.2%); 90–110 average (50%); below 70 intellectually disabled (2.2%).
4. Spearman: g and s by factor analysis. Thurstone: 7 PMAs. Guilford: operations $\times$ contents $\times$ products = 180.
5. Cattell: fluid declines, crystallised grows.
6. Gardner 1983: 8 intelligences, naturalistic added later. Sternberg 1985: analytical, creative, practical.
7. PASS: Das, Naglieri and Kirby (1994), from Luria; CAS for ages 5 to 18.
8. Emotional intelligence: Salovey and Mayer 1990; Goleman 1995; EQ.
9. Twin ladder: 0.90, 0.72, 0.60, 0.50, 0.25.
10. Intellectual disability: IQ about 70 or below, adaptive deficits, developmental onset. Gifted: 130 and above; Renzulli's three rings.
11. Creativity: divergent thinking (Guilford); Torrance's fluency, flexibility, originality, elaboration; threshold near IQ 120.
12. Aptitude is the potential to acquire a skill after training. DAT 1947 has 8 subtests; Ojha's Indian adaptation. GATB has 9 aptitudes.
13. Army Alpha and Beta 1917 were the first group tests; Raven's matrices are culture-reduced; Bhatia's Battery 1955 is India's performance test.
14. Sociometry: Moreno, *Who Shall Survive?* 1934; socio-matrix, sociogram, Northway's target

sociogram.

15. Star = most chosen; isolate = none; mutual pair; clique; three choices on one real criterion.

Sources

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- IGNOU B.Ed. BES-127 *Assessment for Learning*, Block 2 "Techniques and Tools of Assessment" (sociometric technique).
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- National Education Policy 2020: Introduction (fundamental principles); Chapter 4 (support for gifted students/students with special talents).
- Right of Children to Free and Compulsory Education Act, 2009, s.29(2).
- Rights of Persons with Disabilities Act, 2016 (Schedule; ss. 16, 17, 31).
- S.K. Mangal, *Advanced Educational Psychology* (PHI), chapters on intelligence, aptitude, creativity, individual differences and sociometry.
- J.L. Moreno, *Who Shall Survive?* (1934).

Quick facts

- Binet-Simon scale 1905; mental age 1908; Stern's IQ ratio 1912; Terman's Stanford-Binet 1916; Wechsler's deviation IQ (mean 100, SD 15).
- $IQ = MA \div CA \times 100$. MA 12 at CA 10 gives 120; MA 10 at CA 12.5 gives 80.
- NCERT IQ bands: 130+ very superior (2.2%); 120–130 superior (6.7%); 90–110 average (50%); below 70 intellectually disabled (2.2%).
- Spearman: g and s found by factor analysis. Thurstone: 7 primary mental abilities. Guilford: operations $\times$ contents $\times$ products = 180 cells (originally 120).
- Cattell and Horn: fluid intelligence declines with age, crystallised intelligence grows.
- Gardner, Frames of Mind (1983): 8 intelligences in NCERT, naturalistic added later; evidence from case studies, not factor analysis.
- Sternberg's triarchic theory (1985): componential (analytical), experiential (creative), contextual (practical).
- PASS model: Das, Naglieri and Kirby (1994), based on Luria; Cognitive Assessment System for ages 5 to 18.
- Emotional intelligence: term introduced by Salovey and Mayer (1990), popularised by Goleman (1995); expressed as EQ; NCERT calls it the feeling side of intelligence.

- NCERT twin ladder: identical together 0.90, identical apart 0.72, fraternal together 0.60, siblings together 0.50, siblings apart 0.25.
- Intellectual disability = IQ about 70 or below + adaptive behaviour deficits + onset in the developmental period (NCERT 0–18; AAIDD 2021: before 22); one of the 21 RPwD 2016 disabilities.
- Gifted: conventional cut-off IQ 130 and above; Renzulli's three rings are above-average ability, creativity and task commitment.
- Aptitude = potential to acquire a specific skill after training. DAT (1947) has 8 subtests; J.M. Ojha made the Indian adaptation; GATB measures 9 aptitudes.
- Creativity: divergent thinking (Guilford); Torrance scores fluency, flexibility, originality, elaboration; threshold near IQ 120.
- Sociometry: Moreno, Who Shall Survive? (1934). Star = most chosen; isolate = no choices; three choices on one real criterion; Northway's target sociogram.